

SIMATS Engineering

SIMATS Online Programming Lab — Python Programming

Course Name	Python Programming
Course Code	CSA0804
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Topic	CO5-AT2-DEBUGGING AND OPTIMIZATION TASK
Question	Q3 — Broken Group-By Analysis (15 Marks) — Pandas – groupby, agg, sort

Q3. Broken Group-By Analysis [5 Marks] — Pandas – groupby, agg, sort

■ This program contains 4 bug(s). Identify, fix, and optimise.

- (a) Find and fix all errors in the group-by aggregation block.
- (b) Optimise: combine multiple agg functions into one groupby().agg() call.
- (c) Export the result to a new CSV file 'summary.csv' with proper index handling.

Buggy Program:

```
import pandas as pd
df = pd.read_csv('sales.csv')
# Compute total sales and average discount per region
result = df.groupby('Region')[['Sales', 'Discount']].agg({
    'Sales': 'total',
    'Discount': 'mean'
})
result = result.sort_values('Sales', ascending=True)

result.rename(columns={'Sales': 'Total Sales'}, inplace=False)
print(result.head)
result.to_csv('summary.csv', index=True)
```

(a) Errors Identified

Bug 1 – Invalid aggregation function (Line 5):

'Sales': 'total' is not a recognised pandas aggregation name — there is no built-in 'total' function, so **agg()** raises an error. It should be 'Sales': 'sum'.

Bug 2 – rename() result discarded:

result.rename(columns={'Sales': 'Total Sales'}, inplace=False) returns a renamed copy, but with **inplace=False** and no reassignment the original **result** is never updated — the column stays named 'Sales'. It must be reassigned: **result = result.rename(columns={...})**.

Bug 3 – Missing parentheses on head():

print(result.head) prints the bound method object itself, not the first rows of the DataFrame. It should be **print(result.head())**.

Bug 4 – Index not handled before export (part c):

After **groupby('Region')**, 'Region' sits in the DataFrame's index, not as a normal column. Exporting straight to CSV without resetting the index relies on **to_csv** implicitly writing that index; the requested “proper index handling” instead calls for **result.reset_index()** to turn 'Region' into a normal column first, then exporting with **index=False** so there is no duplicated/implicit index column.

(b) Corrected and Optimised Program

The program below fixes all four bugs. The aggregation already uses a single **groupby().agg()** call combining **sum** and **mean** in one pass, so fixing the invalid 'total' key satisfies the optimisation requirement in part (b) without needing separate **groupby()** calls.

```
"""
C05-AT2 - Debugging and Optimisation Task
Challenge 3: Broken Group-By Analysis [Pandas - groupby, agg, sort]
-----
Loads sales data from CSV, computes total sales and average
discount per region in a single groupby().agg() call, sorts
by total sales, and exports the summary to CSV with the
region index handled correctly.
"""

import pandas as pd

# -----
# 1. Load the CSV file
# -----
df = pd.read_csv('sales.csv')

# -----
# 2. Compute total sales and average discount per region
# (fixed: 'total' -> 'sum'; single combined agg() call)
# -----
result = df.groupby('Region')[['Sales', 'Discount']].agg({
    'Sales': 'sum',
    'Discount': 'mean'
})

# -----
# 3. Sort by total sales, ascending
# -----
result = result.sort_values('Sales', ascending=True)

# -----
# 4. Rename the Sales column (fixed: result reassigned)
# -----
result = result.rename(columns={'Sales': 'Total Sales'})

# -----
# 5. Preview the result (fixed: head() called with parentheses)
# -----
print(result.head())

# -----
# 6. Export to CSV with proper index handling (part c):
```

```
# turn the Region index into a normal column, then export
# without writing a duplicate/implicit index column
# -----
result = result.reset_index()
result.to_csv('summary.csv', index=False)
```

(c) Export and Index Handling

result.reset_index() moves 'Region' out of the index and into a normal, labelled column.

result.to_csv('summary.csv', index=False) then writes only the real columns (Region, Total Sales, Discount) without an extra unlabelled index column.

This keeps **summary.csv** clean for downstream use, since every column — including Region — is an explicit named column rather than relying on pandas' implicit index export.

Expected Sample Output

Exact values depend on the contents of **sales.csv**. For a file with columns *Region*, *Sales*, *Discount* covering 4 regions, the output would look like:

	Region	Total Sales	Discount
0	West	18200.0	0.12
1	South	21750.0	0.09
2	North	24600.0	0.15
3	East	27300.0	0.11

(summary.csv written with columns: Region, Total Sales, Discount)

Conclusion

The original group-by block used an aggregation name pandas does not recognise, discarded the result of **rename()**, printed a method reference instead of calling **head()**, and exported the summary without deliberately handling the Region index. The corrected version uses a single **groupby().agg()** call with valid aggregation names ('sum' and 'mean'), reassigns every transformation back to **result**, calls **head()** correctly, and resets the index before exporting with **index=False** so **summary.csv** contains clean, explicitly labelled columns.